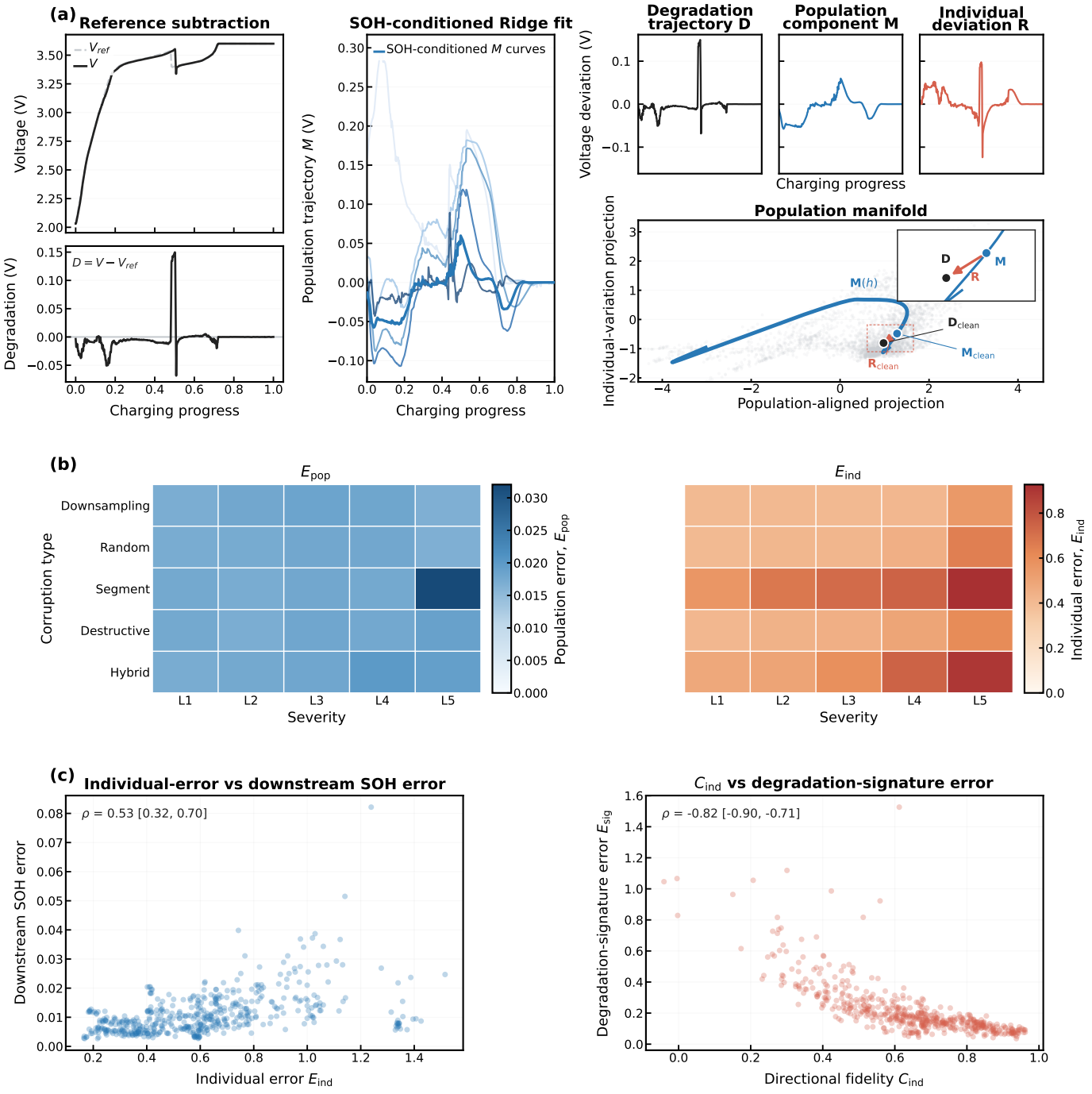

(a)
Reference subtraction
V_{ref}
V
Voltage (V)
$D = V - V_{ref}$
Degradation (V)
Charging progress
SOH-conditioned Ridge fit
SOH-conditioned M curves
Population trajectory M (V)
Charging progress
Degradation trajectory D
Population component M
Individual deviation R
Voltage deviation (V)
Charging progress
Population manifold
M(h)
D
M
R
D_{clean}
M_{clean}
R_{clean}
Individual-variation projection
Population-aligned projection
(b)
E_{pop}
Downsampling
Random
Segment
Destructive
Hybrid
Corruption type
L1
L2
L3
L4
L5
Severity
Population error, E_{pop}
E_{ind}
Severity
Individual error, E_{ind}
(c)
Individual-error vs downstream SOH error
ρ = 0.53 [0.32, 0.70]
Downstream SOH error
Individual error E_{ind}
C_{ind} vs degradation-signature error
ρ = -0.82 [-0.90, -0.71]
Degradation-signature error E_{sig}
Directional fidelity C_{ind}